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Perceptron

We tried a wide range of hyperparameters but removed them from the PDF as it would generate a 50 page report. We tried decreasing the learning rate as well as increasing it by multiplying it by a factor of 10 but the best result we obtained was surprisingly without a learning rate decay and a small learning rate.

To accomplish this, the number of epochs had to be higher.

We observed that the parameters below for the RICE dataset had a better accuracy than the ones provided.

RICE DATASET

Optimal hyperparameters:	learning_rate:0.005, epochs:100 , decay:1 (False)
Training accuracy:	99.95
Validation accuracy:	99.86
Test accuracy:	99.89

Fashion-MNIST DATASET

Optimal hyperparameters:	learning_rate:0.5, epochs:20 , decay:1 (False)
Training accuracy:	84.17
Validation accuracy:	82.32
Test accuracy:	82.18

SVM

We tried a smaller learning rate as well as a couple of regularization constants, but then accomplished the best results for the fashion set and a high decay. When training for the rice dataset, we had to increase the number of epochs, as well as the learning rate and have a smaller decay learning rate.

RICE DATASET

Optimal hyperparameters:	learning_rate: 2.5, epochs:80, regularization_constant: 0.01, decay_rate: 0.95
Training accuracy:	98.82
Validation accuracy:	98.87
Test accuracy:	98.74

Fashion-MNIST DATASET

Optimal hyperparameters:	learning_rate: 0.5, epochs:10, regularization_constant: 0.05, decay_rate: 0.85
Training accuracy:	84.33
Validation accuracy:	83.27
Test accuracy:	82.37

Softmax

For the Softmax regression experiment on the Fashion-MNIST dataset, we experimented with learning rates from 0.0005 to 0.5, epochs between 50 and 100, regularization constants of 0.005, 0.05, and 0.5, decay values of 0.85, 0.9, 0.95, and 1, and batch sizes of 1024, 2048, and 4096. The best-performing model used a learning rate of 0.5, 70 epochs, a regularization constant of 0.05, and a decay value of 0.85, with a mini-batch size of 4096. We also tested learning rate decay, which helped stabilize training and improve generalization.

For the softmax regression on the Rice dataset, we experimented with learning rates ranging from 0.0005 to 0.5, epochs between 100 and 200, regularization constants from 0.005 to 5, decay values of 0.85, 0.9, and 1, and batch sizes of 1024, 2048, and 4096.

RICE DATASET

Optimal hyperparameters:	learning_rate: 0.0005, epochs:110, regularization_constant: 0.05, decay: 0.9, mini-batch-size: 1024
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Training accuracy:	99,12
Validation accuracy:	99,06
Test accuracy:	99.12

Fashion-MNIST DATASET

Optimal hyperparameters:	learning_rate: 0.5, epochs: 70, regularization_constant: 0.05, decay: 0.85, mini-batch-size: 4096
Training accuracy:	85.08
Validation accuracy:	83.63
Test accuracy:	82.79

Logistic

We experimented with learning rates ranging from 0.01 to 0.5 and varied the number of training epochs between 5 and 100. Our results indicated that a learning rate of 0.05 with 5 epochs provided the best balance between convergence speed and accuracy. Higher learning rates led to unstable training, while lower ones required significantly more epochs to achieve similar results. We also tested learning rate decay using a step decay approach, where the learning rate was reduced by a factor of 0.1 after every two epochs, but it did not provide significant improvements in performance. Ultimately, the best-performing model used a fixed learning rate of 0.05, five training epochs, and a decision threshold of 0.5. We attempted to implement mini-batch SGD but encountered issues, so we opted for full-batch SGD instead. Given that the Rice dataset was small, this choice did not significantly impact training speed and still yielded test accuracies that met our expectations.

RICE DATASET

Optimal hyperparameters:	Learning_rate: 0.05, n_epochs = 5, threshold = 0.5
Training accuracy:	98.90
Validation accuracy:	98.35
Test accuracy:	98.62